Race, explain and transfer

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Evaluate a model against measured performance and defend design decisions.

SUCCESS CRITERIA - I can...

1. Report measured distance, time and average speed.
2. Compare actual speed with a stated prediction.
3. Use evidence to explain three physical design choices.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

At unchanged wheel rpm, circumference becomes 6 times as large. Compare ideal travel speed and state one real-world limitation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: average speed = distance/time
relative deviation = |measured - predicted|/|predicted| × 100%. Prediction must be nonzero. Record real race conditions; this compares with a prediction, not symmetric percent difference.

Worked reasoning: A car covers 3 m in 2 s: average speed 1.5 m/s. A complete explanation also uses its circuit, gearing, friction and evidence from repeated runs; one race time does not isolate a cause.

Your turn: Draw an energy-flow diagram from battery to motion and other transfers; annotate it with one actual result, clearly labeled as measured.

Marzano check - circle: ① ② ③

03 Look and explain

Draw or annotate your final car. Label three choices involving wheels, gears, alignment, wiring or mass, then connect each to a physics idea.

Marzano check - circle: ① ② ③

04 Build the solution

A supplied car travels 5 m in 10 s. Find measured speed. Its prediction was .60 m/s. Use the supplied difference .10 m/s to calculate relative deviation=.10÷.60×100%.

Marzano check - circle: ① ② ③

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05 Use the idea

Use your actual race record to defend one successful design choice and identify one revision for a future build. Cite data or a specific observation for each.

Marzano check - circle: 1 2 3

06 Choose a method

A device receives 60 J mechanical energy at its shaft and produces electrical energy at its terminals. Identify it and name one possible loss.

Marzano check - circle: 1 2 3

07 Exit check

Choose one problem on this sheet. Circle your current 1, 2 or 3 self-check and name the next step that would improve your explanation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Propose one redesign and predict two possible effects, including a tradeoff. State the measurement that would support or challenge your prediction.

Marzano check - circle: 1 2 3